

Technology News

Smart cane to help the blind

A 'smart' cane has been developed by the students at Central Michigan University that helps the blind in moving around by themselves easily.

The cane uses radio frequency identification (RFID) to detect obstacles and alert the user on where and how to navigate while walking. It is equipped with an ultrasonic sensor and works in tandem with a navigational system inside a bag worn by the user. The cane requires RFID flags mounted along the path to help the user in reaching his destination.

A speaker on the bag's strap alerts the user of an obstacle coming in his way and tells him where to walk. For people who can't hear, a special glove vibrates different fingertips to provide direction on where to walk.

Robots assist doctors in life-saving operations

Robots are helping doctors to treat soldiers in combat zones from faraway locations. In these operations, the robot is mounted with a high-tech camera and controlled from another region via a laptop and remote joystick. The robot can move easily because of the sensors located on its torso, which help it to identify obstacles. The camera on the robot captures images of the patient from every angle and a zoom option allows doctors a close-up view of their distant patients.

The technology has been incorporated by Dr Kevin Chung, in-charge of the burn intensive care unit at Brooke Army Medical Center in San Antonio, Texas, to treat patients.

Robots could turn dangerous

Robots could be dangerous to human beings. A situation which is now limited to only sci-fi films, such as the *Terminator* series and *The Matrix*, could

become true.

Leading scientists warn that mankind might lose control over computer-based systems that carry out a growing share of society's workload, from waging war to chatting on the phone, and have already reached a level of indestructibility comparable with a cockroach.

Too much time is being spent on developing artificial intelligence and too little on robot safety. "We're rapidly approaching the time when new robots should undergo tests, similar to ethical and clinical trials for new drugs, before they can be introduced," said Alan Winfield, a professor at the University of the West of England.

Smart home control

Schneider Electric has developed an integrated home control solution called 'Wiser Home Control' that interlinks the electrical, multimedia and telecommunications worlds into one solution. Users can set programs according to their preferred living patterns and control home equipment from anywhere at anytime via 3G mobile phones or Web-enabled devices. For example, if you are sitting in the garden and want to listen to music, the control to which is inside your house, simply turn on the audio system with the help of a mobile phone without going indoors. If you are watching a movie at home and feel that the room is too cold, use the TV remote control to adjust the temperature of the AC from the Microsoft Media Center menu on the remote.

Wiser Home Control connects and controls many different devices and technologies and you can preset shortcuts or direct control. You can also use your mobile phone or computer to monitor your home while you are away and reset security systems if

there are false alarms.

Besides controlling, Wiser Home Control can make things happen automatically. When you leave for work, just press the 'goodbye' button and lights will turn off, blinds will close and the music system will switch off automatically.

Electronic cigarettes for combat soldiers

A combat vet has come up with an electronic cigarette that is safer for soldiers to take with them into the battlefield. This electronic cigarette does not burn tobacco or create second-hand smoke that is considered harmful to others as with regular cigarettes. It looks like a real one and is powered by a rechargeable battery connected to a microprocessor that heats a liquid in the simulated filter to produce a vapour when puffed on. The user has a choice to inhale nicotine in a smaller dose than of a tobacco cigarette or none at all. It even recreates an almost odourless smoke that vanishes quickly.

Electronic cigarettes are excellent substitutes for the regular ones for combat soldiers as these are water-resistant, take up less space and do not require a lighter. They even have an LED light on the end that simulates a burning cigarette. The cost of electronic cigarettes is about half that of their tobacco competitors.

Electronic voting machines can be hacked

Computer scientists have demonstrated how criminals can hack electronic voting machines (EVMs) and steal votes using a malicious programming approach. They used return-oriented programming to make an electronic voting machine turn against itself. To take over the machine, they found a flaw in its software that could be